

CSE 400: Fundamentals of Probability in Computing
Milestone 2 Scribe

Date of Submission: February 19thth, 2026 (11:59 PM)

Question-1: What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

Solution: The probabilistic issue, which is discussed within the framework of this project, is the uncertainty quantification of glacier mass balance and its role in the global sea-level rise (SLR).

The system goal is to spread environmental uncertainties using physically based model of energy -mass conversion to estimate probabilistic limits to glacier melting and sea-level rise.

The system is a model of the impact of uncertainty in the climatic variables on the energy of melt and finally global SLR.

Primary Sources of Uncertainty:

- Snowfall Threshold (T_s)
- Precipitation (P)
- Air Temperature (T_a)
- Cloudiness (n)
- Ice Albedo (α)

These uncertainties influence surface melt energy:

$$Q_m = (1 - \alpha)S_{in} + C_0 + C_1T_a$$

which is converted into melt mass:

$$M = \left(\frac{Q_m}{\rho_w L_f} \right) \Delta t$$

and then scaled to global sea-level rise:

$$\Delta SLR = \frac{\text{Mass Loss (Gt)}}{360}$$

Thus, the probabilistic problem is formulated as a multi-stage uncertainty propagation system.

Question-2: Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

Solution: Modeling of Environmental Random Variables

Five important environmental variables in this project are modeled as random variables in order to model climatic uncertainty and measurement variability. All the variables are given a probability distribution that is in agreement with their physical behaviour and evolved using the glacier energy-mass balance framework.

1. Air Temperature T_a — Continuous Random Variable

Air temperature uncertainty is modeled as a Normal distribution:

$$T_a \sim \mathcal{N}(\mu, \sigma^2), \quad \mu = 0, \quad \sigma = \sqrt{0.3^2 + 0.3^2} \approx 0.424^\circ\text{C}.$$

Probability density function:

$$f(T_a \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(T_a - \mu)^2}{2\sigma^2}\right).$$

Chebyshev bound ($k = 3$):

$$P(|T_a - \mu| \geq 3\sigma) \leq \frac{1}{9} \quad \Rightarrow \quad T_a \in [-1.3, 1.3].$$

2. Precipitation P — Continuous Random Variable

Precipitation is modeled using a Gamma distribution:

$$P \sim \text{Gamma}(k, \theta), \quad k = 2.5, \quad \theta = 3.0.$$

Probability density function:

$$f(P \mid k, \theta) = \frac{P^{k-1} e^{-P/\theta}}{\theta^k \Gamma(k)}, \quad P \geq 0.$$

Mean and variance:

$$E[P] = k\theta = 7.5, \quad \text{Var}(P) = k\theta^2 = 22.5.$$

Chebyshev bound ($k = 3$):

$$P \in [0, 21.73] \quad \text{with probability} \geq 0.89.$$

3. Ice Albedo α_i — Continuous Random Variable

Ice albedo is modeled as a truncated Normal distribution:

$$\alpha_i \sim \mathcal{N}(0.34, 0.05^2), \quad 0 \leq \alpha_i \leq 1.$$

Truncated density:

$$f(\alpha_i) = \frac{\phi\left(\frac{\alpha_i - 0.34}{0.05}\right)}{0.05 \left[\Phi\left(\frac{1-0.34}{0.05}\right) - \Phi\left(\frac{0-0.34}{0.05}\right) \right]}, \quad 0 \leq \alpha_i \leq 1.$$

Energy balance coupling:

$$Q_m = (1 - \alpha_i)S_{\text{in}} + C_0 + C_1 T_a.$$

4. Snowfall Threshold T_s — Transition Random Variable

$$T_s \sim \mathcal{N}(\mu_s, \sigma_s^2), \quad \mu_s = 1.0^\circ\text{C}, \quad \sigma_s = 0.5^\circ\text{C}.$$

Snowfall transition probability:

$$P_{\text{snow}}(T) = \Phi\left(\frac{T_s - T}{\sigma_t}\right).$$

Chebyshev bound ($k = 3$):

$$T_s \in [-0.5, 2.5] \quad (\geq 89\% \text{ probability}).$$

5. Cloudiness n — Continuous Random Variable

$$n \sim \mathcal{N}(0.5, 0.2^2).$$

Probability density function:

$$f(n) = \frac{1}{0.2\sqrt{2\pi}} \exp\left(-\frac{(n - 0.5)^2}{2(0.2)^2}\right).$$

Radiative transfer transformation:

$$T_{\text{cl}} = 1.0 - 0.233n - 0.415n^2.$$

Chebyshev bound:

$$n \in [-0.1, 1.1] \quad \Rightarrow \quad \text{after clipping } n \in [0, 1].$$

Probabilistic Assumptions

- All five variables are treated as independent random variables.
- Uncertainty is assumed stationary over the simulation horizon.
- Physical constraints enforced via truncation/clipping:

$$P \geq 0, \quad 0 \leq \alpha_i \leq 1, \quad 0 \leq n \leq 1.$$

- Chebyshev's inequality:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

- Monte Carlo sampling ($N \geq 10,000$) ensures convergence under the Weak Law of Large Numbers.

Question-3: Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

Solution:

Conditional Dependencies in the System

The system contains conditional dependencies arising from physical coupling in the energy–mass balance framework.

Snow–Rain Partition

The snowfall threshold determines the snow–rain fraction:

$$P_{\text{snow}} = P \cdot \Phi\left(\frac{T_s - T}{\sigma_t}\right).$$

Melt Energy Equation

Air temperature and albedo directly influence melt energy:

$$Q_m = (1 - \alpha)S_{\text{in}} + C_0 + C_1T_a.$$

Dependency Structure

- T_s affects snow fraction.
- P influences accumulation and rain-driven melt.
- T_a affects melt energy.
- n modifies radiative transfer.
- α modifies absorbed solar energy.

Modeling Assumption

Primary variables are sampled independently. However, they become conditionally dependent through the coupled energy balance equations.

Question-4: Describe how the current probabilistic model aligns with the project's implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices.

Solution:

Probabilistic Implementation Framework

The probabilistic model is implemented using:

- Monte Carlo simulation ($N = 10,000$ – $100,000$ samples)
- Sampling from defined probability density functions (PDFs)
- Propagation through physical equations
- Construction of PDF and CDF outputs

Theoretical Justification

The Weak Law of Large Numbers ensures convergence:

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

Chebyshev's inequality provides distribution-free probabilistic bounds:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Implementation Assumptions

- Independence of base variables
- Physical clipping for bounded variables
- Linearized energy balance formulation

These assumptions guide the simulation design and performance evaluation.

Question-5: Describe the current state of your project's probabilistic model. Identify the assumptions, components, or relationships that are presently well-defined, and indicate which aspects are expected to evolve or require refinement in subsequent milestones. Briefly justify your reasoning.

Solution:

Current State

- All 5 random variables defined.
- Specified Means and variances.
- Chebyshev bounds derived
- Application of Monte Carlo.
- Chain of energy-mass-SLR constructed.

Well-Defined Components

- Distributional assumptions
- Physical constants
- Concentration bounds

Expected Refinements

- Correlation modeling (e.g., T_a and cloudiness)
- Bayesian estimation on the basis of observations.
- Albedo feedback modeling Nonlinear modeling Nonlinear feedback

Question-6: Identify any unresolved probabilistic questions or ambiguities in the project at this stage, and indicate which role(s) or tasks are responsible for addressing them in the next milestone.

Solution: Unresolved Questions

- Should Gamma parameters be region-specific?
- Is Chebyshev too conservative for Normal variables?
- How sensitive is SLR to snowfall threshold variance?

Responsibilities

- Statistical modeling lead → refine distributions
- Literature review lead → parameter justification

End of Submission